

Alteplase (tPA) Quick-Use Sheet – Acute Ischemic Stroke

Dose | Prep | Admin | PK | Clinical Considerations | Wastage Info

Dosage 0.9 mg/kg (10% of the dose administered as a bolus and the remaining 90% of the dose as an infusion) or 0.09 mg/kg IV push, then 0.81 mg/kg IV infusion

Max total dose: 90 mg

10% given as IV bolus over 1 minute, followed by 90% IV infusion over 60 minutes

Vial Sizes, Reconstitution & Wastage

Vial Size	Diluent Volume	Final Conc.
50 mg	50 mL SWFI	1 mg/mL
100 mg	100 mL SWFI	1 mg/mL

Wastage Key Points:

- Reconstituted solution stable for up to 8 hours at 2–8°C
- Use 50 mg vial when appropriate to minimize waste
- May be eligible for return & reimbursement:
 - o <https://www.activase.com/ais/dosing-and-administration/product-return.html#:~:text=If%20you%20would%20like%20a,are%20prepaid%20by%20the%20Customer.>

Preparation & Administration

Reconstitution:

- Use provided SWFI to achieve 1 mg/mL (50 mg vial with 50 mL; 100 mg with 100 mL)
- Gently swirl; do not shake

Administration:

- IV bolus (10%) over 1 minute
- Remaining 90% as continuous IV infusion over 60 minutes
- Infuse using programmable pump if available

Onset, Duration, and Half-Life

Onset: Rapid, within minutes

Half-life: Initial ~5 min; terminal ~26–40 min

Duration: Short; requires continuous infusion

Clinical Considerations

Guidelines: Alteplase remains the standard thrombolytic for AIS if within 4.5 hours of symptom onset

Contraindications:

- Refer to inclusion/exclusion criteria

Monitoring:

- Neuro checks and vitals per post-thrombolytic protocol
- BP <180/105 mmHg
- No antiplatelets, anticoagulants, or invasive procedures for 24 hours

References:

<https://www.activase.com/ais/dosing-and-administration/reconstituting.html>